Worksheet 52 · Venn Diagrams & Sets IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Use set notation, including union, intersection and complement • Draw and interpret Venn diagrams • Find probabilities from Venn diagrams		
Grade: / 56	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Venn Diagrams & Sets*

Question 1 (3 marks)

$B = \{b, l, u, e\}$

$G = \{g, r, e, y\}$

$W = \{w, h, i, t, e\}$

(a) List all the members of the set

(i) $B \cup G$

(ii) $W \cap G'$

(2)

Serena writes down the statement $B \cap G \cap W = \varnothing$

(b) Is Serena's statement correct?
You must give a reason for your answer.

(1)

Question 2 (4 marks)

$\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
$A = \{2, 3, 5, 7\}$
$B = \{4, 6, 8, 10\}$

(a) Explain why $A \cap B = \varnothing$

(1)

$x \in \mathscr{E}$ and $x \notin A \cup B$

(b) Write down the **two** possible values of x.

(1)

Set C is such that

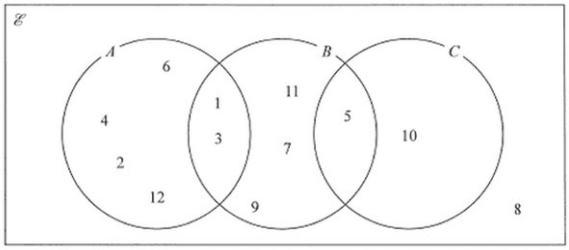
$A \cup B \cup C = \mathscr{E}$
$A \cap C = \{2\}$
$B \cap C' = \{4, 6, 10\}$

(c) List all the members of set C.

(2)

Question 3 (5 marks)

Here is a Venn diagram.



(a) Write down the numbers that are in the set

(i) A

(ii) $B \cup C$

(2)

Brian writes down the statement $A \cap C = \varnothing$

(b) Is Brian's statement correct?
You must give a reason for your answer.

(1)

One of the numbers in the Venn diagram is picked at random.

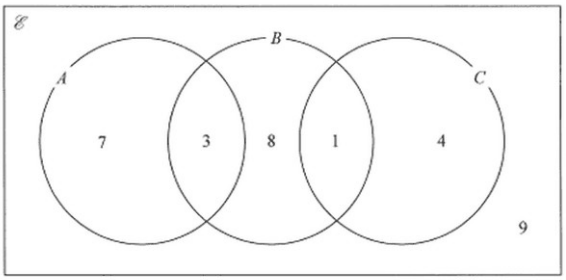
(c) Find the probability that this number is in set C'

(2)

(Total for Question 4 is 5 marks)

Question 4 (3 marks)

The Venn diagram shows a universal set, $\mathscr{E}$, and sets A, B and C.



7, 3, 8, 1, 4 and 9 represent the **numbers** of elements.

Find

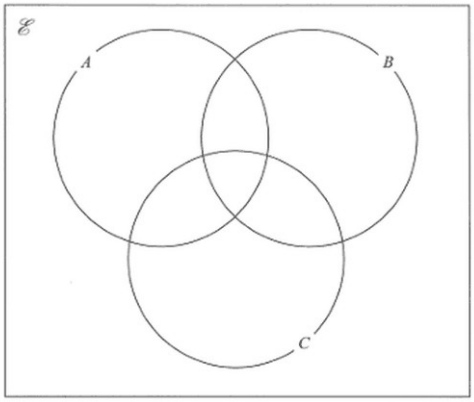
(i) $n(A \cup B)$

(ii) $n(A' \cap C)$

(iii) $n(A' \cup B')$

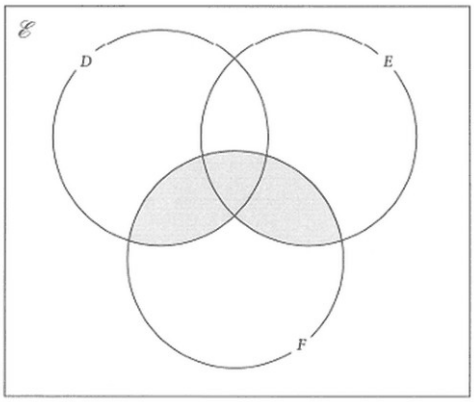
Question 5 (2 marks)

(a) On the Venn diagram, shade the set $(A \cup B)' \cap C$



(1)

(b) Use set notation to describe the shaded region in the Venn diagram below.



(1)

Question 6 (3 marks)

$\mathscr{E}$ = {letters of the alphabet}
B = {b, r, a, z, i, l}
I = {i, r, e, l, a, n, d}

(a) List the members of the set

(i) $B \cup I$

(ii) $B \cap I'$

(2)

K = {k, e, n, y, a}

Cody writes down the statement $B \cap K = \varnothing$
Cody's statement is wrong.

(b) Explain why.

(1)

Question 7 (5 marks)

$\mathscr{E}$ = {23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34}
A = {even numbers}
B = {23, 29, 31}
C = {multiples of 3}

(a) List the members of the set

(i) $B \cup C$

(1)

(ii) $A' \cap C$

(1)

(b) Is it true that $B \cap C = \varnothing$?

Tick (✓) one of the boxes below.

Yes ☐ No ☐

Give a reason for your answer.

(1)

The set D has 4 members and is such that $D \cap (A \cup C) = \varnothing$

(c) List the members of set D

(2)

Question 8 (5 marks)

$\mathscr{E} = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$
$A = \{\text{multiples of } 3\}$
$B = \{\text{odd numbers}\}$

(a) List the members of the set

(i) $A \cap B$

.....

(1)

(ii) $A \cup B$

.....

(1)

(b) Is it true that $24 \in A$?

Tick one of the boxes below.

Yes ☐ No ☐

Give a reason for your answer.

.....

(1)

Set C has 4 members such that $C \cap B' = \{10, 18\}$

(c) List the members of one possible set C

(2)

Question 9 (4 marks)

$\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
$A = \{\text{odd numbers}\}$
$A \cap B = \{1, 3\}$
$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 9, 11, 12\}$

Draw a Venn diagram to show this information.

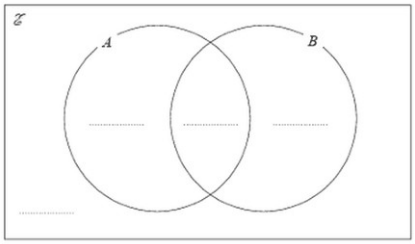


Question 10 (5 marks)

Two events A and B are such that $n(A) = 62$ $n(B) = 30$ and $n(A \cup B) = 68$

Given that $n(\mathscr{E}) = 80$

(a) complete the Venn diagram to show the number of elements in each region.



(2)

An element is chosen at random from $\mathscr{E}$.

(b) Using the Venn diagram, find the probability that this element is in

(i) $A \cap B$

.....

(1)

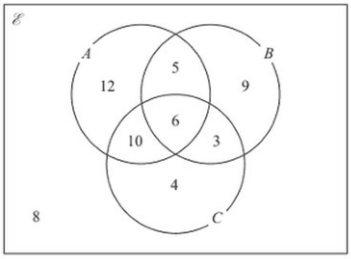
(ii) $A \cup B'$

.....

(2)

Question 11 (3 marks)

The Venn diagram shows a universal set, $\mathscr{E}$ and sets A, B and C.



12, 5, 9, 10, 6, 3, 4 and 8 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....

(1)

(ii) $n(A' \cap B')$

.....

(1)

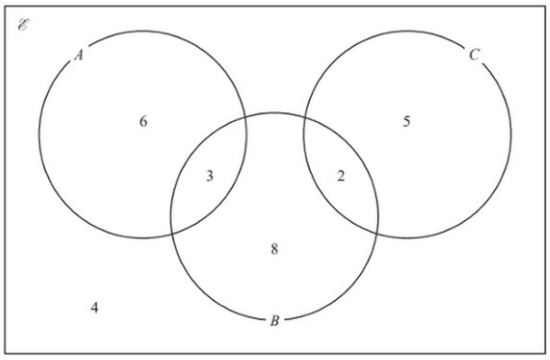
(iii) $n([A \cap B] \cup C)$

.....

(1)

Question 12 (4 marks)

The Venn diagram shows a universal set $\mathscr{E}$ and three sets A, B and C.



6, 3, 8, 2, 5 and 4 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....
(1)

(ii) $n(A \cap C)$

.....
(1)

(iii) $n(B \cap C')$

.....
(1)

(iv) $n(A' \cup B' \cup C')$

.....
(1)

Question 13 (5 marks)

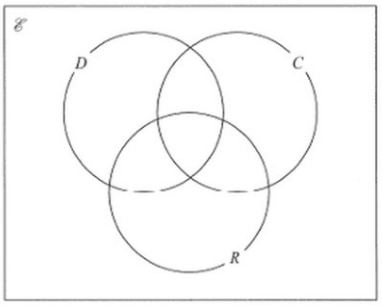
Some students in a school were asked the following question.

"Do you have a dog (D), a cat (C) or a rabbit (R)?"

Of these students

28 have a dog
18 have a cat
20 have a rabbit
8 have both a cat and a rabbit
9 have both a dog and a rabbit
x have both a dog and a cat
6 have a dog, a cat and a rabbit
5 have not got a dog or a cat or a rabbit

(a) Using this information, complete the Venn diagram to show the number of students in each appropriate subset.
Give the numbers in terms of x where necessary.



(3)

Given that a total of 50 students answered the question,

(b) work out the value of x.

$x =$ .....
(2)

Question 14 (5 marks)

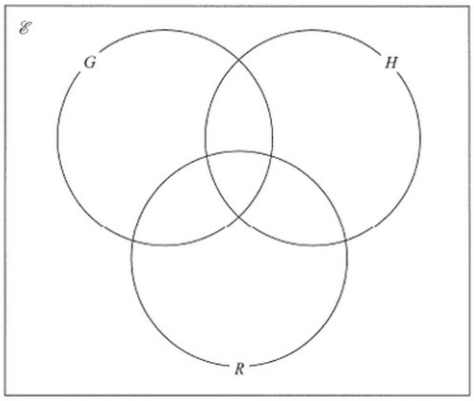
All the students in Year 11 at a school must study at least one of Geography (G), History (H) and Religious Studies (R).

In Year 11 there are 65 students.

Of these students

15 study Geography, History and Religious Studies
21 study Geography and History
16 study Geography and Religious Studies
30 study Geography
18 study only Religious Studies
37 study Religious Studies

(a) Using this information, complete the Venn diagram to show the number of students in each region of the Venn diagram.



(3)

A student in Year 11 who studies both History and Religious Studies is chosen at random.

(b) Work out the probability that this student does **not** study Geography.

(2)